

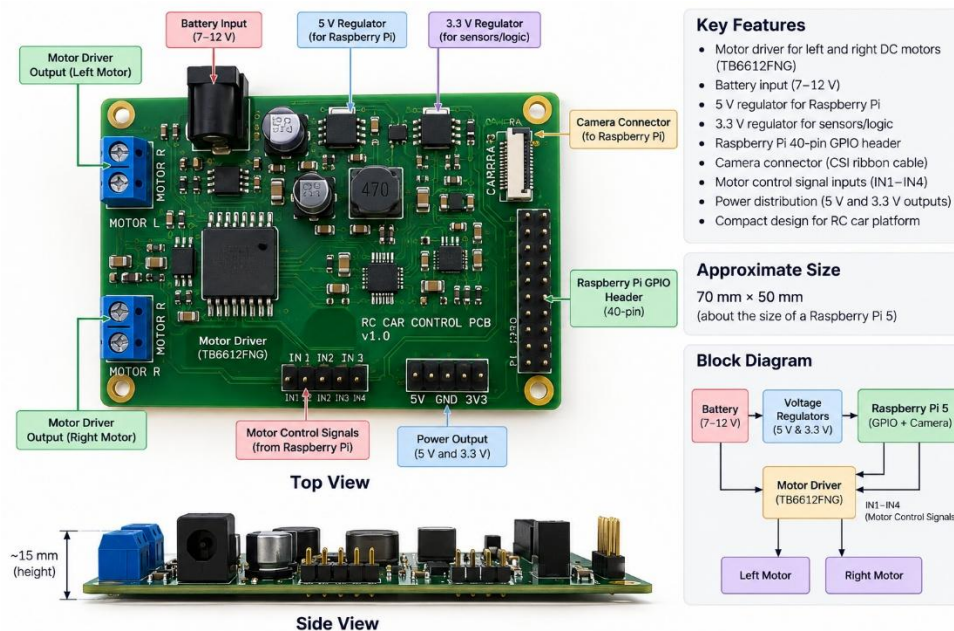
Camera Navigation

Description: The camera view will be divided into two sections, with each section performing a different task. The top half of the camera view will be used for traffic sign recognition, allowing the vehicle to identify signs and respond accordingly. The bottom half will be processed using OpenCV for path detection and navigation.

Main Components:

- **Raspberry Pi 5** — main computer; runs OpenCV, path detection, and traffic-sign recognition.
- **Raspberry Pi Camera** — provides video feed that is split between sign recognition and path detection.
- **DC Gear Motors ×2** — drive the vehicle.
- **Motor Driver** — allows the Raspberry Pi to control motor direction and speed
- **Battery / Power Supply** — supplies power to the motors, Raspberry Pi, and PCB.
- **Custom PCB** — handles power distribution, voltage regulation, motor-driver connections, and other connections between the Raspberry Pi and vehicle hardware.
 - o **Voltage Regulation** — likely incorporated into the custom PCB to provide the correct voltage for the Raspberry Pi and other electronics.

PCB Design:



Pros	Cons
Less Sensors – Camera does both nav and recog	Lighting Conditions
Simpler PCB design	Higher processing demand – could cause delay
“Realistic” Navigation	Camera loss affects both nav and recog
Software Troubleshooting instead of redesigning	More Software development
Lower Cost	

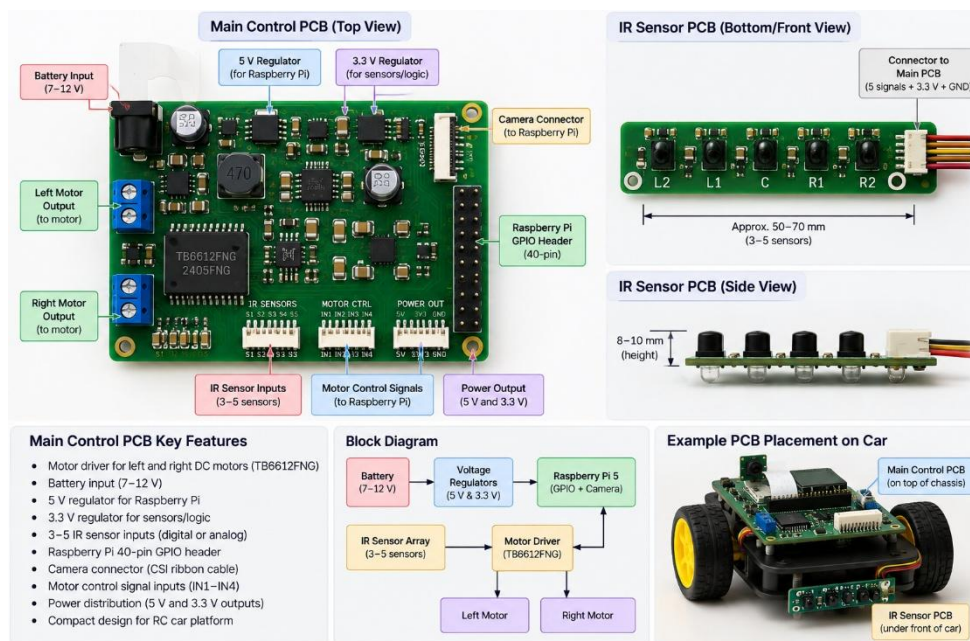
IR Line-Following Navigation

Description: 3–5 IR sensors mounted underneath the front of the car to detect and follow a line on the course. The sensors will determine whether the vehicle is centered on the line or drifting to the left or right. The Raspberry Pi will use these readings to adjust the left and right motor speeds and keep the vehicle on the desired path

Main Components:

- **Raspberry Pi 5**
- **Raspberry Pi Camera** — sign recognition
- **3–5 IR reflectance sensors** — line detection
- **DC Gear Motors ×2**
- **Motor Driver**
- **Battery / Power Supply**
- **Custom PCB**
 - **Voltage Regulation**

PCB Design:



Pros	Cons
Simple line following	Needs clear line for navigation
Multiple sensors = better position detecting	Better contrast between line and surface
Less computation demanding than OpenCV	Sensors placement above lines and spaced out
Camera focus on sign recognition	Ambient lighting could affect contrast
Inexpensive Sensor	Less flexible if course changes (Modular course idea)

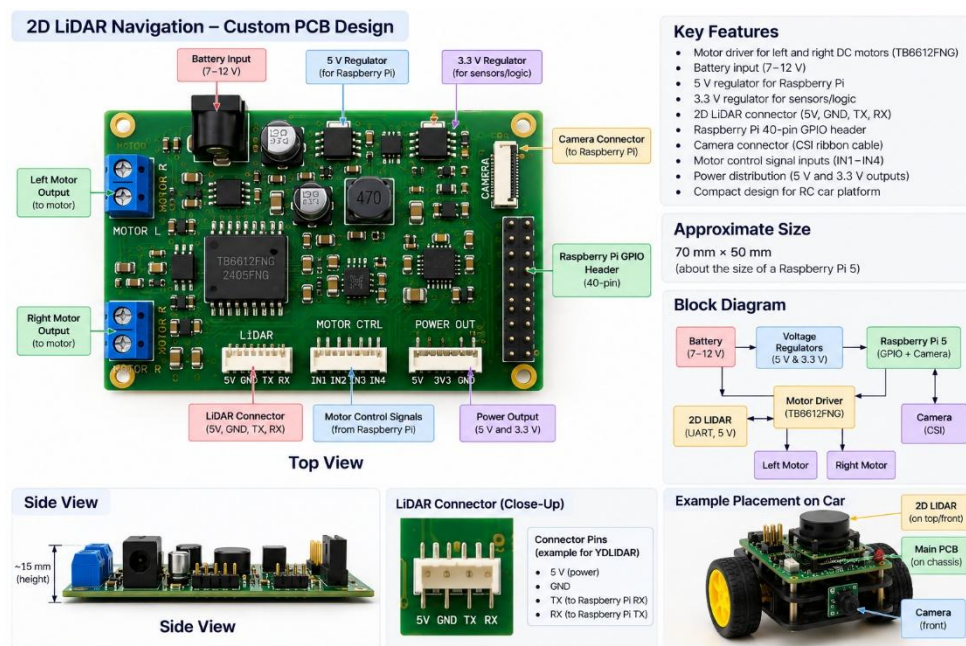
2D Lidar Navigation

Description: 2D LiDAR sensor to measure distances to objects and boundaries around the car. The LiDAR continuously scans the surrounding area and sends distance measurements to the Raspberry Pi. These measurements can be used to determine the vehicle's position relative to walls or other course boundaries and make steering corrections.

Main Components:

- **Raspberry Pi 5**
- **Raspberry Pi Camera**
- **2D LiDAR Sensor** — scans the surroundings and provides distance measurements
- **DC Gear Motors ×2**
- **Motor Driver**
- **Battery / Power Supply**
- **Custom PCB**
 - **Voltage Regulation**

PCB Design:



Pros	Cons
360° view around the vehicle	More Expensive
Does not require a physical line for navigation	More Complex Navigation Software
Less affected by lighting conditions	additional weight and power consumption
Detects walls, obstacles, and course boundaries	LiDAR data increases the workload on Pi
	More complex course building

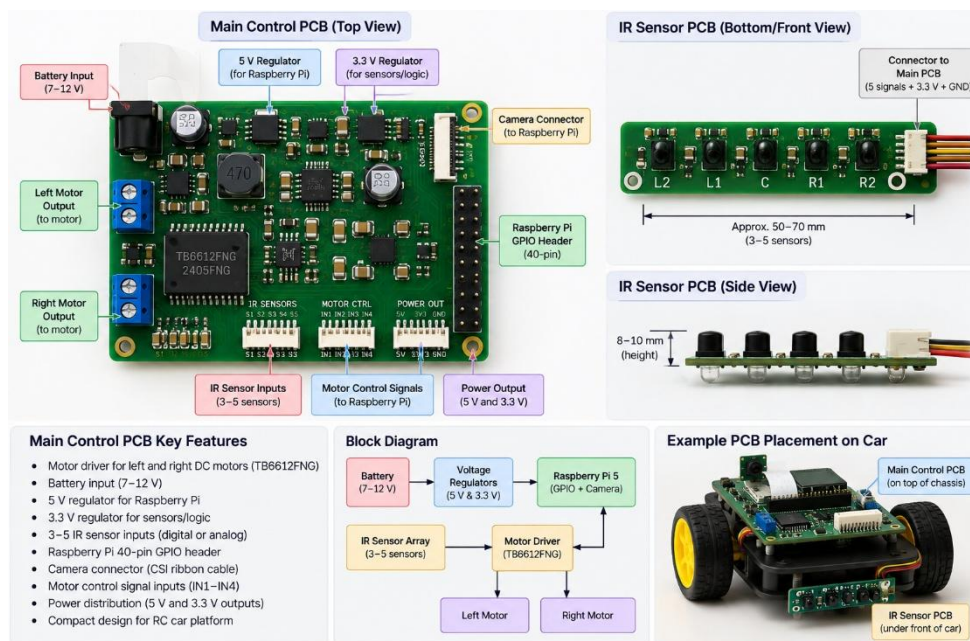
Camera + IR Line-Following Navigation

Description: Camera and 3–5 IR sensors for navigation. The IR sensors mounted underneath the front of the vehicle will detect the line and provide fast feedback for basic line-following corrections. The camera will use OpenCV to provide additional information about the path ahead while also performing traffic sign recognition

Main Components:

- **Raspberry Pi 5**
- **Raspberry Pi Camera** — sign recognition and Path information
- **3–5 IR reflectance sensors** — line detection
- **DC Gear Motors ×2**
- **Motor Driver**
- **Battery / Power Supply**
- **Custom PCB**
 - o **Voltage Regulation**

PCB Design: Same as IR Navigation



Pros	Cons
Provides redundancy between two	More Complex software, info from 2 navigation systems
Combines the reliability of IR line following with the flexibility of camera navigation.	Lighting could affect both camera and line contrast
Camera can detect information farther ahead than the IR sensors	how much control each navigation system has when the camera and IR sensors disagree?
IR sensors provide fast feedback for steering corrections.	

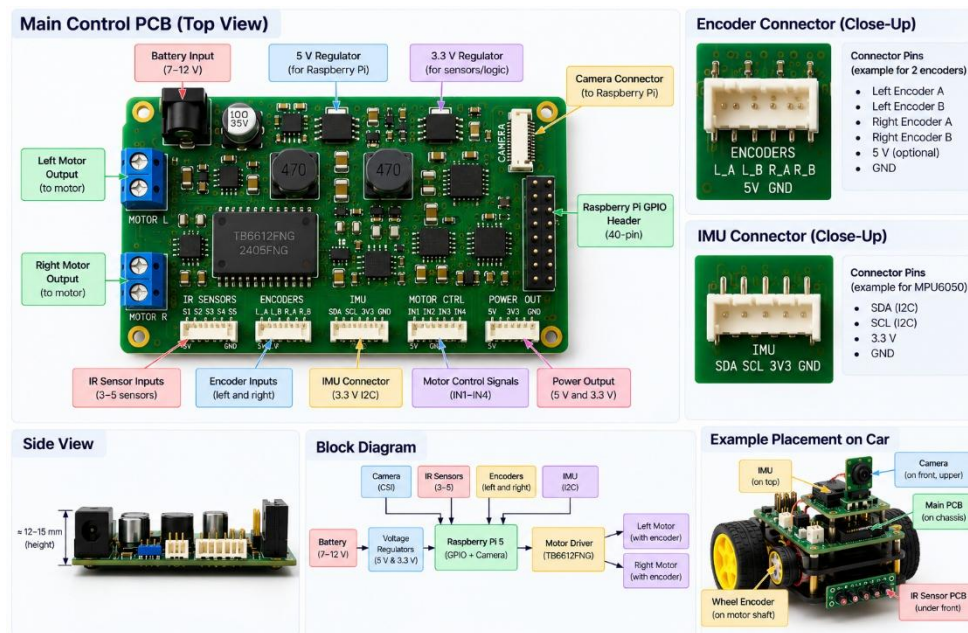
Camera + IR + Encoder + IMU Navigation

Description: Camera, 3–5 IR sensors, wheel encoders, and an IMU for navigation. The IR sensors will provide reliable line-following information, while the camera uses OpenCV for additional path detection and traffic sign recognition. Wheel encoders will measure the rotation of the wheels to estimate the vehicle's speed and distance traveled. The IMU will measure the vehicle's movement and orientation, helping detect turns and changes in direction.

Main Components:

- **Raspberry Pi 5**
- **Raspberry Pi Camera** — sign recognition and Path information
- **3–5 IR reflectance sensors** — line detection
- **2× Wheel Encoders** — measure left and right wheel rotation
- **IMU (Accelerometer + Gyroscope)** measures movement and orientation
- **DC Gear Motors ×2**
- **Motor Driver**
- **Battery / Power Supply**
- **Custom PCB**
 - **Voltage Regulation**

PCB Design: Same as IR Navigation



Pros	Cons
provide more navigation information	More complicated PCB design
make controlled and repeatable turns	Software more complicated because data from multiple sensors
redundancy if one navigation method becomes unreliable	increased Pi processing
	increase power consumption.

